

Boundary Compensation – Compensated Islands VIII: Finite-Resolution Atlas Metrics and Certified Local Distances

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Abstract

BC-CI-VI introduced certified reachability on a finite diagnostic graph. BC-CI-VII measured the multiplicity of robust support behind such reachability claims by certification entropies. The present paper develops the eighth module of the Compensated Islands branch. It defines finite-resolution distance diagnostics on certified reachability atlases. The phrase “metric” is used only in the finite graph and audit-diagnostic sense. The construction does not define a spacetime metric, Lorentzian interval, physical distance, proper time, geodesic, curvature tensor, or metric tensor on hidden reality.

Let $\mathcal{G} = (V, E)$ be a finite directed diagnostic graph with a robustly certified edge set E_{rob} inherited from BC-CI-VI and with certification-support data inherited from BC-CI-VII. A certified local distance protocol is a declared package

$$\mathcal{D} = (E_{\text{rob}}, L, c, m, \sigma, \mathcal{R}),$$

where L is a path budget, $c : E_{\text{rob}} \rightarrow \mathbb{R}_{\geq 0}$ is a declared edge audit cost, $m : E_{\text{rob}} \rightarrow \mathbb{R}_{\geq 0}$ is a declared margin or stability score, σ is a declared support-size or entropy diagnostic, and $\mathcal{R}$ is the reset-barrier rule. From this data the paper defines several layer-aware quantities: certified depth distance, additive audit distance, bottleneck margin distance, entropy-support distance, and quotient-level symmetrized distances.

The main thesis is that there is no single universal distance on a certified atlas. Depth, cost, margin, entropy support, reset obstruction, and quotient comparison measure different layers. Some of these diagnostics satisfy ordinary triangle inequalities under reset-free concatenation and additivity assumptions; others satisfy only weakened audit inequalities or no metric law at all. This distinction is the contribution of the module: it converts the dangerous phrase “emergent metric” into a controlled finite-dimensional status tree for certified local distance comparison.

The theorem-level results are deliberately modest: budget-depth and additive audit distances are extended directed pseudometrics on reset-free certified components under declared concatenation rules; bottleneck diagnostics are monotone under path improvement but not additive distances; entropy-support distances need not satisfy a triangle inequality unless an explicit concatenation-support lower bound is declared; and all distance statuses are stable only away from reset, margin, and threshold walls. The construction is a finite-resolution certification audit, not spacetime geometry.

Keywords. Boundary Compensation; compensated islands; certified local distances; finite-resolution metrics; reachability atlas; certification entropy; graph pseudometric; bottleneck margins; reset barriers; claim hygiene.

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Version note v0.1.1. This clean revision incorporates the three-review audit of v0.1.0. It corrects minor typographical and formatting artifacts, standardizes the symmetrized-distance notation, clarifies that the protocol components c , m , and σ define independent layer-aware diagnostics, strengthens the bottleneck and reset-barrier warnings, and adds a mesh-refinement caution. No new physical or theorem-level claims are introduced. All distance-like terminology remains constrained by the non-import rule: certified distance diagnostics are not physical distances or spacetime geometry.

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Notation table

Symbol	Meaning
$\mathcal{G} = (V, E)$	finite directed diagnostic graph
E_{rob}	robustly certified edge set from BC-CI-VI
$p \preceq_{\text{cert}} q$	certified reachability relation
$\mathcal{P}_L^{\text{rob}}(p, q)$	robust certified paths from p to q with path budget L
L	declared path budget
$c(e)$	declared nonnegative edge audit cost
$m(e)$	declared edge margin or stability score
$\sigma(p, q)$	declared support-size, entropy, or weight-volume diagnostic
$\mathcal{R}$	reset-barrier rule
$d_{\text{depth}}^{(L)}$	minimum certified path length under budget L
d_c	additive audit distance induced by declared edge costs
$\beta(p, q)$	best bottleneck margin between p and q
d_{bott}	bottleneck-derived margin-risk diagnostic, not generally additive
d_S	entropy-support distance, typically $-\log$ of a declared support fraction
$\sim_{\text{cert}}$	mutual certified reachability equivalence
$V/\sim_{\text{cert}}$	quotient partial-order base from BC-CI-VI
CERTIFIED_DISTANCE	status for a valid declared distance diagnostic
RESET_BLOCKED	undefined certified continuation under the declared atlas protocol
ENTROPY_DISTANCE_NONMETRIC	status for entropy diagnostic not satisfying metric laws
METRIC_TENSOR_NONCLAIM	explicit non-claim status excluding metric-tensor interpretation

1 Introduction: from support multiplicity to local distance diagnostics

BC-CI-VI constructed a finite directed reachability layer. Given diagnostic points $p, q \in V$, the relation

$$p \preceq_{\text{cert}} q \quad (1)$$

means that a reset-free certified path from p to q exists, supported by robust hidden-section data inherited from BC-CI-V. It is a preorder after the length-zero path convention and concatenation of certified paths. It is not physical causality.

BC-CI-VII then observed that reachability may be nonunique. A single relation $p \preceq_{\text{cert}} q$ can be supported by multiple robust paths, multiple hidden-section classes, multiple weight-policy regions, or multiple macrostates. This led to certification entropies such as

$$S_{\text{path}}^{(L)}(p, q) = \log |\mathcal{P}_L^{\text{rob}}(p, q)| \quad (2)$$

and weight-volume diagnostics such as

$$S_W(p, q) = \log \frac{\mu(\mathcal{W}_{\varepsilon}^{\text{rob}}(p, q))}{\mu(\mathcal{W}_{\text{adm}})}. \quad (3)$$

These quantities measure epistemic breadth of certified support, not thermodynamic entropy.

The present module asks the next finite-resolution question. If one certified relation is supported broadly and another narrowly, if one path has high margins and another crosses low-margin edges, or if two nodes are connected only through many reset-risk regions, how should

such situations be compared? The answer cannot be a single universal metric. Different layers generate different distance diagnostics.

The guiding transition is

$$\text{certified reachability} + \text{support multiplicity} \longrightarrow \text{certified local distance diagnostics.} \quad (4)$$

This transition is mathematical and audit-theoretic. It is not a derivation of spacetime geometry.

2 Non-import rule: distance diagnostics are not geometry

This paper uses distance terminology in a restricted sense. A certified distance is a function or status table used to compare nodes, paths, supports, margins, or quotient components in a finite diagnostic graph.

The following exclusions are part of the definition of the module:

$$d_{\text{cert}} \neq \text{spacetime interval}, \quad d_{\text{cert}} \neq \text{proper time}, \quad d_{\text{cert}} \neq \text{physical length}, \quad (5)$$

$$\text{certified path} \neq \text{geodesic}, \quad \text{entropy gradient} \neq \text{force}, \quad \text{atlas metric} \neq \text{Lorentzian or Riemannian geometry} \quad (6)$$

The status

$$\text{METRIC_TENSOR_NONCLAIM} \quad (7)$$

is assigned to any reading that treats the distance diagnostics of this paper as a physical metric tensor.

3 Recall of BC-CI-VI and BC-CI-VII

3.1 Certified reachability

Let $\mathcal{G} = (V, E)$ be a finite directed graph whose vertices are diagnostic points or chart-level states. BC-CI-VI declares a robust edge set

$$E_{\text{rob}} \subseteq E. \quad (8)$$

A path $\rho = (e_1, \dots, e_k)$ is robustly certified if every edge belongs to E_{rob} and no reset barrier is crossed. The certified reachability relation is

$$p \preceq_{\text{cert}} q \iff \exists \rho \in \mathcal{P}^{\text{rob}}(p, q). \quad (9)$$

3.2 Certification entropy

BC-CI-VII introduces finite path budgets L and finite support sets

$$\mathcal{P}_L^{\text{rob}}(p, q), \quad \mathcal{X}_L^{\text{rob}}(p, q), \quad \mathcal{W}_\varepsilon^{\text{rob}}(p, q). \quad (10)$$

The corresponding entropies depend on the declared support protocol. No entropy value is meaningful before the support class and counting or measure protocol are declared.

3.3 Why distance is the next layer

Reachability is binary; entropy measures multiplicity. Distance diagnostics compare routes, margins, and support geometry. For example, two reachable pairs may have the same reachability status but differ by minimum path depth, total audit cost, bottleneck margin, or weight-support fraction. This is the precise scope of BC-CI-VIII.

4 Certified local distance protocols

Definition 4.1 (Certified local distance protocol). *A certified local distance protocol on a finite diagnostic graph is a declared tuple*

$$D = (V, E_{\text{rob}}, L, c, m, \sigma, \mathcal{R}), \quad (11)$$

where V is the node set, E_{rob} is the robustly certified edge set, $L \in \mathbb{N} \cup \{\infty\}$ is a path budget, $c : E_{\text{rob}} \rightarrow \mathbb{R}_{\geq 0}$ is a nonnegative edge audit cost, $m : E_{\text{rob}} \rightarrow \mathbb{R}_{\geq 0}$ is a margin or stability score, σ is a declared support-size or entropy diagnostic, and $\mathcal{R}$ is the reset-barrier rule. The components c , m , and σ define independent layer-aware diagnostics: a protocol may use any one of them, or a declared combination of them, provided the chosen layer and aggregation rule are fixed before the audit.

Remark 4.2 (Protocol-first convention). *The protocol must be declared before distance values are computed. Retrofitting c , m , σ , L , or $\mathcal{R}$ after inspecting the desired distance status is a tuning artifact. The correct status is then*

$$\text{DISTANCE_TUNING_ARTIFACT}. \quad (12)$$

Definition 4.3 (Certified path family). *For $p, q \in V$, define*

$$\mathcal{P}_L^{\text{rob}}(p, q) = \{\rho : p \rightarrow q ; \rho \text{ uses only } E_{\text{rob}}, |\rho| \leq L, \rho \text{ crosses no reset barrier}\}. \quad (13)$$

If this set is empty, the pair receives the status

$$\text{UNREACHABLE} \quad (14)$$

relative to the declared protocol.

5 Depth distance

Definition 5.1 (Certified depth distance). *The certified depth distance is*

$$d_{\text{depth}}^{(L)}(p, q) = \inf\{|\rho| : \rho \in \mathcal{P}_L^{\text{rob}}(p, q)\}, \quad (15)$$

with the convention that the infimum of the empty set is ∞ and $d_{\text{depth}}^{(L)}(p, p) = 0$ by the length-zero path convention.

Theorem 5.2 (Depth triangle inequality under budget closure). *Assume reset-free concatenation is allowed and the budget is closed under concatenation: if $\rho_1 \in \mathcal{P}_{L_1}^{\text{rob}}(p, q)$ and $\rho_2 \in \mathcal{P}_{L_2}^{\text{rob}}(q, r)$, then $\rho_2 \circ \rho_1 \in \mathcal{P}_{L_1+L_2}^{\text{rob}}(p, r)$. Then*

$$d_{\text{depth}}^{(L_1+L_2)}(p, r) \leq d_{\text{depth}}^{(L_1)}(p, q) + d_{\text{depth}}^{(L_2)}(q, r). \quad (16)$$

In the unbudgeted finite graph setting with reset-free concatenation, d_{depth} is an extended directed pseudometric on each certified component.

Proof. Choose paths whose lengths realize the finite minima, or approximate the infima in the extended notation. Concatenating them gives a certified path from p to r with length equal to the sum of the two lengths. Taking the infimum over all such paths gives the inequality. Reflexivity follows from the length-zero path. Directional symmetry is not assumed. $\square$

Remark 5.3 (Directed, not physical). *The word “distance” here means minimum graph budget. It is directed and may be infinite. It is not a physical travel time, spatial distance, or signal propagation duration.*

6 Additive audit distance

Definition 6.1 (Declared edge audit cost). *Let $c : E_{\text{rob}} \rightarrow \mathbb{R}_{\geq 0}$ be a declared nonnegative function. For a certified path $\rho = (e_1, \dots, e_k)$ define*

$$C(\rho) = \sum_{j=1}^k c(e_j). \quad (17)$$

The additive audit distance is

$$d_c(p, q) = \inf\{C(\rho) : \rho \in \mathcal{P}^{\text{rob}}(p, q)\}. \quad (18)$$

If no robust path exists, $d_c(p, q) = \infty$.

Theorem 6.2 (Additive audit pseudometric law). *If $c(e) \geq 0$ and reset-free certified paths are closed under concatenation, then*

$$d_c(p, r) \leq d_c(p, q) + d_c(q, r) \quad (19)$$

for all $p, q, r \in V$, with the usual extended-real convention. Also $d_c(p, p) = 0$.

Proof. The proof is the same as for depth distance, replacing path length by the additive cost $C(\rho)$. Nonnegativity gives $d_c(p, p) = 0$ through the empty path and prevents negative-cost loops from collapsing the audit. $\square$

Remark 6.3 (Cost is not action). *The edge audit cost c is a certification-policy diagnostic. It is not an action, not an energy, not a Lagrangian, and not a Hamiltonian. If the cost is chosen after the desired shortest path is known, the status is `DISTANCE_TUNING_ARTIFACT`.*

7 Bottleneck margin diagnostics

Definition 7.1 (Path bottleneck margin). *Let $m : E_{\text{rob}} \rightarrow \mathbb{R}_{\geq 0}$ be a declared edge margin. The bottleneck margin of a path is*

$$M(\rho) = \min_{e \in \rho} m(e), \quad (20)$$

with M of the empty path treated as ∞ by convention. The best bottleneck margin from p to q is

$$\beta(p, q) = \sup\{M(\rho) : \rho \in \mathcal{P}^{\text{rob}}(p, q)\}. \quad (21)$$

Proposition 7.2 (Bottleneck improvement monotonicity). *Adding a robust certified path from p to q cannot decrease $\beta(p, q)$. Increasing an edge margin cannot decrease any $\beta(p, q)$.*

Proof. Both operations enlarge the set over which a supremum is taken or increase values inside that set. Therefore the supremum cannot decrease. $\square$

Remark 7.3 (Bottleneck is not additive). *A bottleneck diagnostic generally does not satisfy an additive triangle inequality. One may define $d_{\text{bott}}(p, q) = 1/\beta(p, q)$ when $\beta(p, q) > 0$, but this is a margin-risk diagnostic for the weakest certified edge along a route, not a distance in the ordinary additive sense and not automatically a metric. The correct status is*

$$\text{BOTTLENECK_DIAGNOSTIC_NONADDITIVE} \quad (22)$$

unless additional assumptions are declared.

8 Entropy-support distances

Definition 8.1 (Support fraction). *Assume a finite measure μ is declared on a support space and let $F(p, q) \in [0, 1]$ be a declared support fraction for robust reachability from p to q . For example,*

$$F_W(p, q) = \frac{\mu(\mathcal{W}_\varepsilon^{\text{rob}}(p, q))}{\mu(\mathcal{W}_{\text{adm}})}. \quad (23)$$

The entropy-support distance is

$$d_S(p, q) = \begin{cases} -\log F(p, q), & F(p, q) > 0, \\ \infty, & F(p, q) = 0. \end{cases} \quad (24)$$

Remark 8.2 (Base of logarithm). *Throughout this section $\log$ denotes the natural logarithm, or any fixed logarithm base $b > 1$. The choice of base only rescales distance values and does not affect ordering or statuses.*

Theorem 8.3 (Entropy distance needs a declared composition law). *The function $d_S = -\log F$ need not satisfy a triangle inequality. If, however, a declared support-composition lower bound holds,*

$$F(p, r) \geq C F(p, q) F(q, r) \quad (25)$$

for a fixed $C > 0$, then

$$d_S(p, r) \leq d_S(p, q) + d_S(q, r) - \log C. \quad (26)$$

In particular, if $C \geq 1$, then d_S satisfies the ordinary directed triangle inequality on that triple.

Proof. Taking negative logarithms gives

$$-\log F(p, r) \leq -\log C - \log F(p, q) - \log F(q, r), \quad (27)$$

which is the stated inequality. $\square$

Remark 8.4 (No entropy-force reading). *The quantity d_S measures lack of declared support fraction. A gradient of d_S on a graph is an audit diagnostic. It is not force, proper time, geodesic deviation, thermodynamic potential, or physical curvature.*

9 Quotient and symmetrized distances

BC-CI-VI replaces a preorder by a quotient partial order using mutual reachability:

$$p \sim_{\text{cert}} q \iff p \preceq_{\text{cert}} q \text{ and } q \preceq_{\text{cert}} p. \quad (28)$$

Distance diagnostics can be lifted to quotient components, but the quotient convention must be declared.

Definition 9.1 (Symmetrized component distance). *Given a directed extended distance d , define*

$$d_{\max}^{\text{sym}}(p, q) = \max\{d(p, q), d(q, p)\} \quad (29)$$

or

$$d_+^{\text{sym}}(p, q) = d(p, q) + d(q, p), \quad (30)$$

where finite. On quotient components $[p]$, define component distances by taking either an infimum, supremum, or declared representative rule over $p \in [p]$, $q \in [q]$.

Proposition 9.2 (Symmetrization gives a pseudometric under finiteness). *If d is an extended directed pseudometric and both directed distances are finite on a component, then $d_{\max}^{\text{sym}}$ and d_+^{sym} are symmetric pseudometrics on that component.*

Proof. Nonnegativity and symmetry are immediate. The triangle inequality follows from the directed triangle inequality applied in both directions, then taking either the maximum or the sum. $\square$

Remark 9.3 (Zero-distance classes). *A pseudometric becomes a metric only after quotienting by zero distance. This is a mathematical quotient convention, not identification of physical points in spacetime.*

10 Atlas-local distances and chart transitions

A certified atlas may contain multiple charts with local edge sets, local costs, and local margins. A distance value computed in one chart need not be silently continued through a wall or reset zone.

Definition 10.1 (Chart-local distance). *Let $\mathcal{A}_i$ be a certified chart with local robust edge set $E_{\text{rob},i}$ and local protocol D_i . A chart-local distance is any distance diagnostic computed using only the declared data of D_i .*

Definition 10.2 (Transition-compatible distance comparison). *On an overlap of charts $\mathcal{A}_i$ and $\mathcal{A}_j$, distance comparison is transition-compatible if the overlap transition certificate is valid, reset barriers are absent, edge labels are matched by the declared transition map, and the distance protocol transforms by a declared comparison rule.*

Theorem 10.3 (No silent metric continuation). *If a chart transition fails its certification predicate or crosses a reset barrier, then a local distance diagnostic from one chart cannot be silently continued into the other chart. The correct status is*

$$\text{CERTIFICATE_RESET} \tag{31}$$

or

$$\text{RESET_BLOCKED}. \tag{32}$$

Proof. A chart-local distance is defined only relative to its declared certified data. A failed overlap certificate removes the declared comparison rule. Therefore continuation would use data outside the protocol. This is exactly the no-silent-continuation rule from the atlas modules. $\square$

Remark 10.4 (Reset-blocked does not mean infinite distance). *The status `RESET_BLOCKED` is not an assignment of an infinite physical distance and not a horizon-like event. It means that certified continuation is undefined under the declared atlas protocol because the required transition or reset-free certificate is absent.*

11 Stability under perturbation and refinement

Definition 11.1 (Distance-stability margin). *A distance value is margin-stable at tolerance η if all edge certificates, reset-free statuses, and cost/margin values used by its minimizing or selected path remain valid under the declared deterministic perturbation class.*

Proposition 11.2 (Stability away from distance walls). *Suppose every edge used by all selected minimizing paths has certificate margin greater than η and every excluded reset edge remains reset-blocked with margin greater than η . If the edge costs change by at most δ_c per edge under perturbation and selected paths have length at most L , then the additive audit distance changes by at most $L\delta_c$, provided no new lower-cost certified path enters the admissible set.*

Proof. The existing selected path remains certified by the margin assumption. Its cost changes by at most $L\delta_c$. If no new lower-cost certified path enters, the optimum can move only within this declared bound. The condition explicitly excludes wall crossing. $\square$

Remark 11.3 (Refinement caution). *Certified local distances are protocol- and mesh-relative diagnostics. Mesh refinement can add nodes, split edges, reveal reset barriers, change path multiplicity, or alter the selected distance layer. Distance comparison across refinements requires an explicit coarsening or embedding map. Without it, the status is*

$$\text{REFINEMENT_COMPARISON_UNDECLARED.} \quad (33)$$

12 Minimal examples

Example 12.1 (Depth differs from entropy support). *Let $A \rightarrow B \rightarrow D$ and $A \rightarrow C \rightarrow D$ be two robust paths of length two. The depth distance from A to D is two. The path-count entropy with budget two is $\log 2$. Depth and support multiplicity measure different layers.*

Example 12.2 (A low-cost but fragile edge). *Suppose $A \rightarrow D$ is a direct edge with cost 1 but margin 0.01, while $A \rightarrow B \rightarrow D$ has total cost 3 and margins 0.5. Additive audit distance selects the direct path; bottleneck margin flags it as fragile. The correct report is not one scalar metric but a status vector.*

Example 12.3 (Entropy distance is not automatically metric). *Let support fractions be $F(A, B) = 0.9$, $F(B, C) = 0.9$, and $F(A, C) = 0.1$. Then*

$$-\log F(A, C) > -\log F(A, B) - \log F(B, C). \quad (34)$$

The entropy-support distance violates the ordinary triangle inequality unless a support-composition law is declared.

Example 12.4 (Reset wall blocks comparison). *If every path from p to q crosses a reset edge, then p and q may be graph-connected but not certified-distance connected. The status is `RESET_BLOCKED`, not a large physical distance.*

13 Computational supplement specification

The companion supplement for this module should compute only finite graph diagnostics. A minimal implementation should return:

Output	Meaning
<code>depth_distance_matrix</code>	minimum robust path length under budget
<code>cost_distance_matrix</code>	additive audit distance using declared costs
<code>bottleneck_margin_matrix</code>	best minimum edge margin along robust paths
<code>entropy_support_distance</code>	entropy-support distance ($-\log$ of support-fraction) table, where declared
<code>reset_barrier_report</code>	blocked pairs and reset edges
<code>status_table</code>	layer-aware statuses for each pair
<code>quotient_component_distances</code>	optional distances between mutual-reachability components

The supplement must not label these outputs as spacetime metrics, geodesic lengths, causal intervals, or curvature tensors.

14 Controlled claims

The controlled claims of this paper are:

- (i) certified reachability can be enriched by layer-aware distance diagnostics;
- (ii) depth and additive audit distances satisfy directed triangle inequalities under reset-free concatenation and nonnegative declared costs;
- (iii) bottleneck margins are useful stability diagnostics but are not additive distances without further assumptions;
- (iv) entropy-support distances are not automatically metrics and require explicit support-composition assumptions for triangle-type inequalities;
- (v) atlas-local distance comparison requires valid certified transitions;
- (vi) distance statuses are stable only away from reset, threshold, and margin walls.

15 Non-claims and limitations

This paper does not claim:

- (i) a spacetime metric;
- (ii) a Lorentzian or Riemannian manifold;
- (iii) a metric tensor born from certificates;
- (iv) physical distance, proper time, geodesics, or light cones;
- (v) curvature, gravity, field equations, or continuum geometry;
- (vi) a causal-set theorem;
- (vii) empirical prediction or physical signal propagation;
- (viii) a path integral, action principle, energy functional, Hamiltonian, or Lagrangian;
- (ix) uniqueness of hidden reality or hidden-sector reconstruction.

The phrase “metric-like” is allowed only as shorthand for finite graph distance diagnostics under declared certification rules.

16 Non-Claim Ontological Reading

One may read this module as a candidate bridge from finite-resolution reachability and support multiplicity toward comparison structures that resemble distance at the audit level. This reading is not a theorem of physical geometry. It says only that once a finite diagnostic graph has certified edges, margins, costs, supports, and reset barriers, one can compare nodes by several controlled distance-like quantities.

The safe summary is

$$\text{certified distance diagnostics, not spacetime geometry.} \tag{35}$$

17 Outlook: after local distances

A later module may consolidate the Compensated Islands chain into an engineering guide or overview:

$$\text{residual layers} \rightarrow \text{transport} \rightarrow \text{atlases} \rightarrow \text{selection} \rightarrow \text{robustness} \quad (36)$$

$$\rightarrow \text{reachability} \rightarrow \text{certification entropy} \rightarrow \text{certified local distances}. \quad (37)$$

Such a synthesis should preserve the same non-import rule. No step in this chain by itself derives physical spacetime, thermodynamics, dynamics, or gravity.

18 Publication-control audit

This clean manuscript includes title, author, ORCID, version, date, abstract, keywords, author responsibility note, AI-use note, notation table, numbered equations, theorem/proposition numbering, controlled claims, non-claims, and a computational supplement specification. The v0.1.1 clean release has undergone the requested typographical and formatting corrections, status-string consistency audit, supplement-language check, PDF compilation, and rendered-page visual inspection.

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